

Requirements_News_Equity_Research_Tool

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Tags	

The goal is to provide an OOTB solution for Equity Researchers to improve their decision efficiency.

Problem statement:

1. Huge amount of information is generated about the target company on a daily basis which can have impact on the stock or can be helpful in stock projection.
2. It is humANELY impossible to go through a number of articles about different target companies and draw a conclusion about them based on all of this daily.
3. Losing out on reacting to this info in time can cause major profit leakages for an Equity researcher.

Problems with existing solution (ChatGPT, etc.):

1. ChatGPT is trained on data till August 2025, while the need is to extract information from near real time released articles.
2. ChatGPT is not able to go and visit to given website links and extract information.
3. Copy pasting full articles in input text to ChatGPT is a tedious task, also, in addition there is a token limit of around 4000 tokens that can be pasted on the ChatGPT interface input text box at once, and articles can on average breach this token limit.

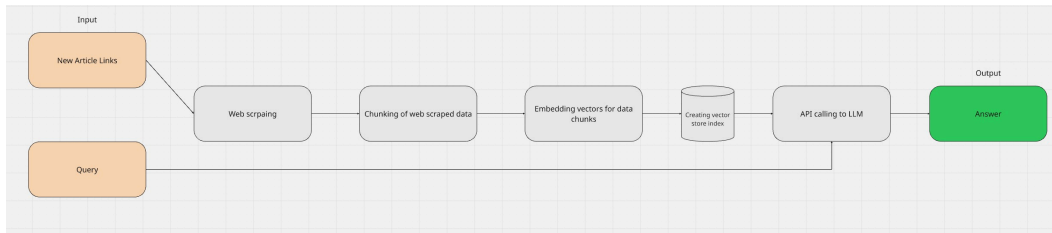
Proposed solution:

An OOTB chat application tool that can take these recently released article links about target companies from the user, process them and retrieve the info user asked for.

Business value:

1. To be able to retrieve relevant information for news articles quickly and efficiently. (Example, summary or some specific information).
2. To be able to retrieve collated information scattered among multiple articles with their source links (Could Have Requirement).
3. Faster information gathering is key for reacting quickly to the news and increasing the probability of maximizing profits.
4. Leads to data-driven decision making rather than human intuition, minimizing the human error in the process.

Architecture:



Requirements:

User Story	Acceptance Criteria	Notes
As a Data Scientist , I should be able to web-scrape user-input article link.	1. (Must Have) System accepts a valid URL input from UI 2. (Must Have) To be able to extract the paragraph text from the given article. 3. (Must Have) To be able to extract the table data from the given article. 4. (Should Have) Handle empty URLs. 5. (Should Have) Handle invalid URLs. 6. (Could Have) Handle multiple URLs at the same time	Number of URLs must be at least 1 and at most 3
As a Data Scientist , I should be able to pre-process web-scraped input dataset.	1. (Must Have) Tables are converted into structured text format 2. (Must Have) Final output is normalized into a consistent document schema 3. (Must Have) Metadata is attached to each document	
As a Data Scientist , I should be able to split the dataset into chunks for downstream processing	1. (Must Have) To be able to decide on the chunking strategy. 2. (Must Have) To derive optimal chunk size for the use case. 3. (Must Have) To derive optimal chunk overlap for the use case. 4. (Must Have) No chunk exceeds model token limits 5. (Could Have) To have configurable chunk size for the use case. 6. (Could Have) To have	

User Story	Acceptance Criteria	Notes
	configurable chunk overlap for the use case.	
As a Data Scientist , I should be able to generate vector embeddings from chunks	1. (Must Have) To be able to decide the embedding to be used for the use case. 2. (Must Have) To be able to decide the vector index to store the vector embeddings. 3. (Could Have) To decide on the token limit for each embedding 4. (Should Have) Handle API failure	
As a Data Scientist , I should be able to store embeddings in a vector database	1. (Must Have) Decide upon the optimal vector index to be used for the use case 2. (Must Have) All embeddings are stored with associated metadata 3. (Must Have) Storage is efficient and scalable for expected dataset size 4. (Could Have) Handle multiple article vectors efficiently	Number of URLs must be at least 1 and at most 3
As a Data Scientist , I should be able to retrieve the relevant context and provide answer to the query grounded in retrieved context	1. (Must Have) To decide the retrieval strategy from the vector store index. 2. (Must Have) To decide the optimal top-k chunks value for the use case 3. (Could Have) To have top-k chunks configurable by the user 4. (Must Have) To decide the LLM to be used. 5. (Must Have) To decide the prompt to be used for LLM Chain. 6. (Should Have) Handle API failure 7. (Could Have) Provide source link in answer when multiple articles are listed by the user.	Number of URLs must be at least 1 and at most 3
As a UI/UX engineer , I should be able to facilitate user to provide input article link and query .	1. (Must Have) UI contains input field for URL Link 2. (Must Have) UI contains input field for user query 3. (Must Have) UI has a button to trigger processing 4. (Must Have) UI displays generated answer clearly 5. (Should Have) UI shows loading indicator during processing	Number of URLs must be at least 1 and at most 3

User Story	Acceptance Criteria	Notes
	6. (Could Have) UI contains multiple input fields for URL Links	
As a User , I should get a fast and reliable response	1. (Must Have) Results are reproducible for same input 2. (Must Have) End-to-end response time is within acceptable limits 3. (Should Have) System does not crash on invalid inputs 4. (Could Have) Able to retrieve information scattered among different articles with their source links	1. Number of URLs must be at least 1 and at most 3 2. Acceptable time limit is 5-40 seconds